

Name: Maths Class:

SYDNEY TECHNICAL HIGH SCHOOL



Year 12 Mathematics

HSC Course

Assessment 2

March, 2015

Time allowed: 70 minutes

General Instructions:

- Marks for each question are indicated on the question.
- Approved calculators may be used
- All necessary working should be shown
- Full marks may not be awarded for careless work or illegible writing
- ***Begin each question on a new page***
- Write using black or blue pen
- All answers are to be in the writing booklet provided
- A set of Standard Integrals is provided at the rear of this Question Booklet, and may be removed at any time.

Multiple Choice

Questions 1-5

5 Marks

Questions 6-10

45 Marks

Question 1

Which expression is used to find the interest earned when \$ P is invested for n years and interest of 10% p.a. is compounded twice yearly?

- A. $I = P(1.1)^n - P$
- B. $I = P(1.05)^n - P$
- C. $I = P(1.05)^{2n} - P$
- D. $I = P(1.05)^{n/2} - P$

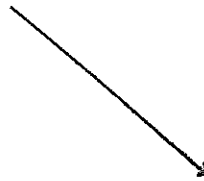
Question 2

Which decreasing function shows positive second derivative for all x values?

A.



C.



B.



D.



Question 3

For which x values is the curve $y = \frac{1}{3}x^3 - 4x + 5$ increasing?

- A. $-2 < x < 2$
- B. $x < -2$ and $x > 2$
- C. $x > 2$
- D. $x < -2$

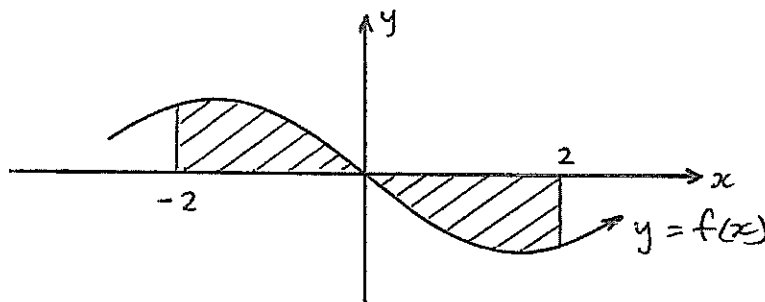
Question 4

Identify which type of stationary point occurs at (3,2) on a curve where $\frac{dy}{dx} = (6 - 2x)^2(x + 5)\sqrt{x}$.

- A. maximum turning point
- B. minimum turning point
- C. horizontal point of inflexion on a rising curve
- D. horizontal point of inflexion on a falling curve

Question 5

Which expression does not correctly evaluate the shaded area for the odd function below?

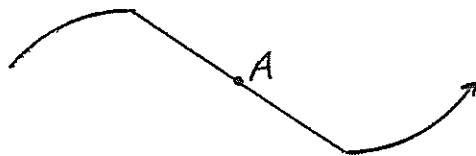


- A. $-2 \int_0^2 f(x) dx$
- B. $-2 \int_0^{-2} f(x) dx$
- C. $2 \int_{-2}^0 f(x) dx$
- D. $2 \int_0^2 f(x) dx$

Question 6 (8 marks) Start on a new page.

a) Select two statements that are true for the point A shown on the function below?

1



- 1. $f'(x) < 0$
- 2. $f'(x) > 0$
- 3. $f'(x) = 0$
- 4. $f''(x) < 0$
- 5. $f''(x) > 0$
- 6. $f''(x) = 0$

b) A curve has gradient function $x + 5$. If the curve passes through $(4,0)$:

i) find the equation of the curve.

1

ii) find the equation of the tangent to the curve at this point.

1

c) Find :

i) $\int (2x + 5)^6 dx$

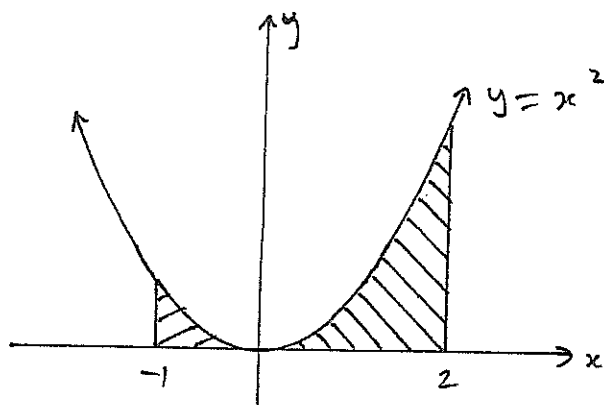
1

ii) $\int \frac{(x-1)(x+1)}{x^2} dx$

2

d) Find the exact shaded area below :

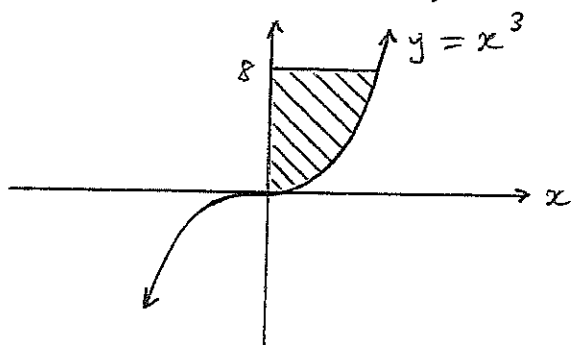
2



Question 7 (10 marks) Start on a new page.

a) Find the shaded area with reference to the y-axis.

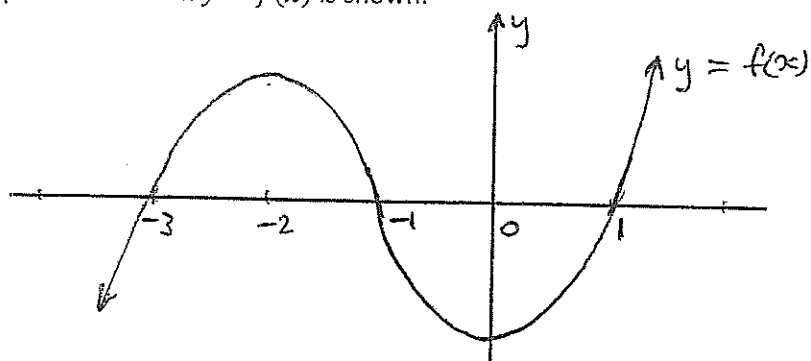
3



b) Find the shaded area in b) above using an alternative method.

2

c) The graph of a function $y = f(x)$ is shown.



i) For which x values does the curve have :

$\alpha) f'(x) > 0 ?$

1

$\beta) f''(x) < 0 ?$

1

ii) On a separate diagram, neatly sketch a possible graph of $y = f'(x)$.

2

Show relevant x values.

iii) On a separate diagram, neatly sketch a possible graph of $y = f''(x)$.

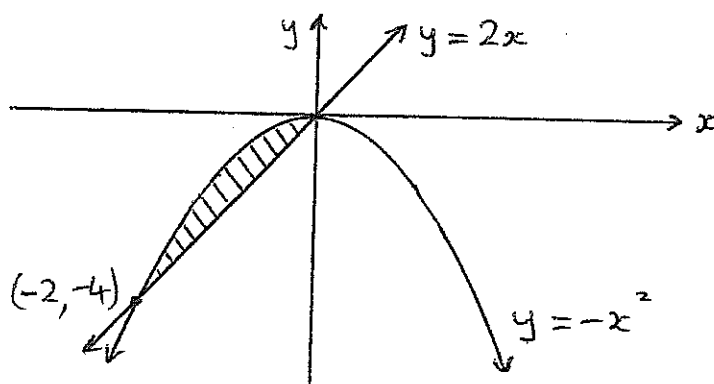
1

Show relevant x values.

Question 8 (9 marks) Start on a new page.

a) Find the shaded area between the two curves as shown.

3



b) Twenty-five kangaroos were released on an island. The population P of kangaroos on the island t years after release is given by $P = -t^3 + 6t^2 + 25$, for $0 \leq t \leq 6$.

i) After how many years was the population a maximum?

2

ii) Determine that the graph of P has a point of inflexion.

2

iii) Sketch the curve $P = -t^3 + 6t^2 + 25$, for $0 \leq t \leq 6$.

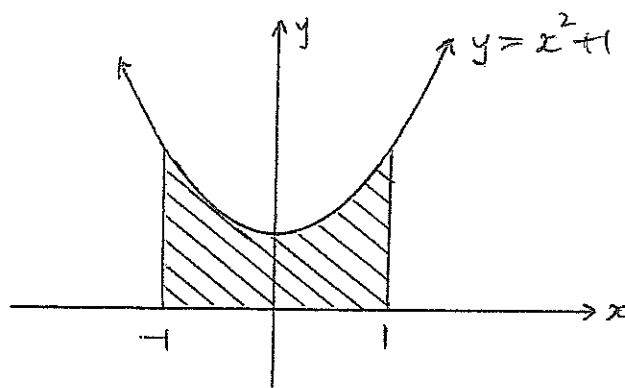
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Question 9 (8 marks) Start on a new page.

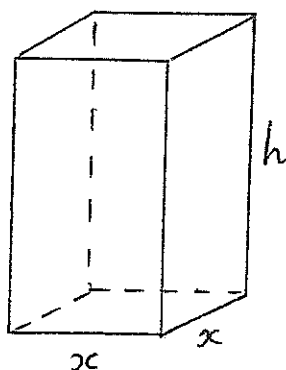
- a) The shaded area below is rotated about the x -axis.

3

Find, in terms of π , the volume of the generated solid.



- b) A closed square prism box has width x cm, length x cm and height h cm as shown.



The sum of width + length + height is to remain constant at 20 cm.

- i) Express h in terms of x .

1

- ii) Show that the box has surface area A given by $A = 80x - 6x^2$

1

- iii) Find, and prove, the maximum surface area of the box.

3

Question 10 (10 marks) Start on a new page.

- a) The curve $y = ax^3 + x^2 + bx$ has a horizontal point of inflexion when $x = 1$.

2

Find the values of a and b .

b) A new car has a cash price of \$50,000 but is bought using time payment.

The interest rate is 9% p.a. and monthly repayments R are made for 4 years.

A repayment R is made immediately after that month's interest is charged.

Let A_n be the amount still owing on the loan after n months.

- | | |
|--|---|
| i) Write a simple expression for A_1 | 1 |
| ii) Show that $A_2 = 50000(1.0075)^2 - R(1 + 1.0075)$ | 1 |
| iii) Write an expression for A_{48} . Simplify your answer. | 2 |
| iv) Show that $R = \$1244$ (to the nearest dollar). | 1 |
| v) Find the amount still owing after 24 months (to the nearest dollar). | 1 |
| vi) After 24 months, the car buyer increases her repayments to \$2200 per month in order to pay off the loan more quickly. | 2 |

Using v) above, find how many remaining months n are now required to pay off the loan. Give your answer to the nearest whole value.

END OF TEST

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

24 SOLUTIONS (2015 - TASK 2)

① C ② A ③ B ④ C ⑤ D

⑥ a) 1 and 6

b) i) $y = \frac{x^2}{2} + 5x + c$

$$x=4, y=0 \Rightarrow 0 = 8 + 20 + c$$

$$c = -28$$

$$\therefore y = \frac{x^2}{2} + 5x - 28$$

ii) $m_T = 9 \Rightarrow$ tangent is $y - 0 = 9(x - 4)$

$$\therefore y = 9x - 36$$

c) i) $\frac{(2x+5)^7}{14} + c$

ii) $\int \frac{x^2-1}{x^2} dx = \int (1 - x^{-2}) dx$

$$= x + \frac{1}{x} + c$$

d) $A = \int_{-1}^2 x^2 dx$

$$= \left[\frac{x^3}{3} \right]_{-1}^2$$

$$= \frac{8}{3} - \left(-\frac{1}{3} \right)$$

$$= 3$$

⑦ a) $x = y^{\frac{1}{3}}$

$$A = \int_0^8 y^{\frac{1}{3}} dy$$

$$= \left[\frac{3}{4} y^{\frac{4}{3}} \right]_0^8$$

$$= \frac{3}{4} (8^{\frac{4}{3}} - 0)$$

$$= \frac{3}{4} \times 16$$

$$= 12 u^2$$

b)

$$A = (8 \times 2) - \int_0^2 x^3 dx$$

$$= 16 - \left[\frac{x^4}{4} \right]_0^2$$

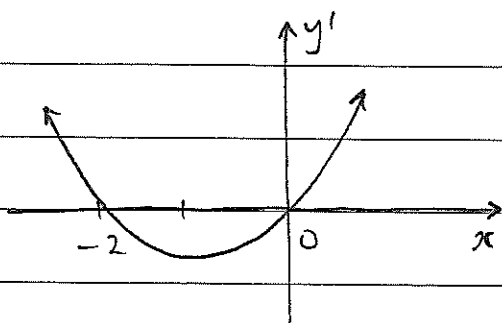
$$= 16 - (4 - 0)$$

$$= 12 u^2$$

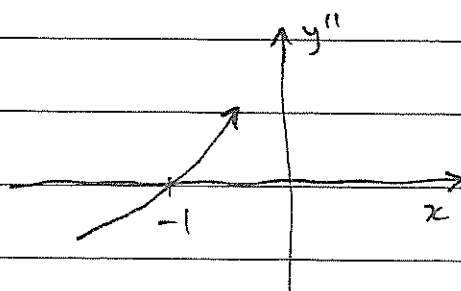
c) i) 2) $x < -2, x > 0$

β) $x < -1$

ii)



iii)



⑧

a) $A = \left| \int_{-2}^0 (2x + x^2) dx \right|$

$$= \left| \left[x^2 + \frac{x^3}{3} \right]_{-2}^0 \right|$$

$$= \left| 0 + 0 - \left(4 - \frac{8}{3} \right) \right|$$

$$= \frac{1}{3} u^2$$

b) i) max P when $\frac{dP}{dt} = 0$

$$\frac{dP}{dt} = -3t^2 + 12t = 0$$

$$-3t(t-4) = 0$$

$$\therefore t = 0, 4$$

Test: $P'' = -6t + 12$

When $t = 0, P'' > 0$

When $t = 4, P'' < 0$

$$\therefore \text{max. P. when } t = 4 \text{ years}$$

ii) P. of I. when $P'' = 0$ and

changes sign

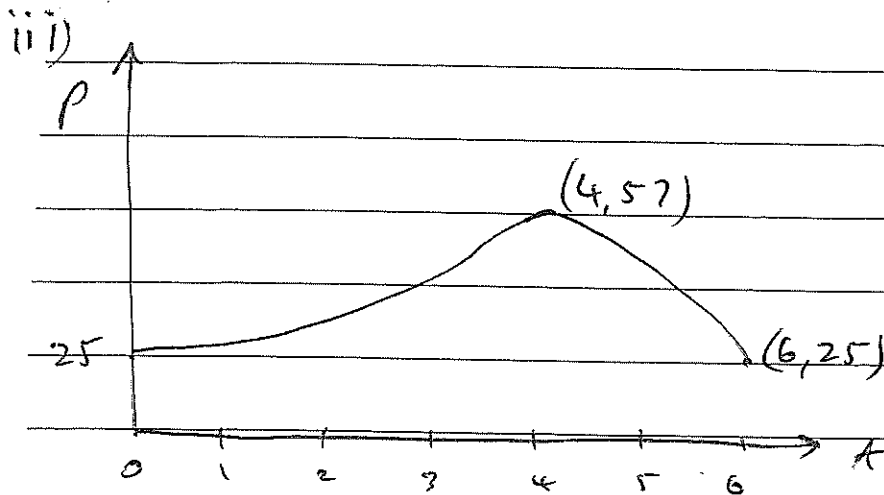
$$P'' = -6t + 12 = 0$$

$$\therefore t = 2$$

t	1.9	2	2.1
P''	+	0	-

$\therefore$ concavity changes

$$\therefore \text{P. of I. when } t = 2 \text{ years}$$



9 a)

$$V = 2\pi \int_0^1 (x^2 + 1)^2 dx$$

$$= 2\pi \int_0^1 (x^4 + 2x^2 + 1) dx$$

$$= 2\pi \left[\frac{x^5}{5} + \frac{2x^3}{3} + x \right]_0^1$$

$$= 2\pi \left[\left(\frac{1}{5} + \frac{2}{3} + 1 \right) - 0 \right]$$

$$= \frac{56\pi}{15} u^3$$

b) i) $x + x + h = 20$

$$\therefore h = 20 - 2x$$

ii) $A = 2x^2 + 4xh$

$$= 2x^2 + 4x(20 - 2x)$$

$$= 2x^2 + 80x - 8x^2$$

$$= 80x - 6x^2 \text{ as reqd.}$$

iii) Minimum A when $\frac{dA}{dx} = 0$

$$\frac{dA}{dx} = 80 - 12x = 0$$

$$\therefore x = 6\frac{2}{3}$$

and prove max : $A'' = -12 (< 0)$

$\therefore$ max. Area is proved for $x = 6^{2/3}$

$$\text{and } A_{\max} = 80 \times 6^{2/3} - 6 \times (6^{2/3})^2 \\ = 266^{2/3} u^2$$

(10) a) Given $f'(1) = 0$ and $f''(1) = 0$ ①

$$\text{Now, } f'(x) = 3ax^2 + 2x + b \Rightarrow 3a + 2 + b = 0$$

$$f''(x) = 6ax + 2 \Rightarrow 6a + 2 = 0$$

$$\therefore a = -\frac{1}{3}$$

$$\text{Sub in ①, } \therefore b = -1$$

$$\text{b) i) } A_1 = 50000 \times 1.0075 - R$$

$$\text{ii) } A_2 = A_1 \times 1.0075 - R$$

$$= (50000 \times 1.0075 - R) \times 1.0075 - R$$

$$= 50000 \times 1.0075^2 - 1.0075R - R$$

$$= 50000 \times 1.0075^2 - R(1 + 1.0075) \text{ as reqd.}$$

$$\text{iii) } A_{48} = 50000 \times 1.0075^{48} - R(1 + 1.0075 + 1.0075^2 + \dots + 1.0075^{47}) \\ = 50000 \times 1.0075^{48} - R \frac{(1.0075^{48} - 1)}{0.0075}$$

$$\text{iv) } A_{48} = 0 \Rightarrow \frac{R(1.0075^{48} - 1)}{0.0075} = 50000 \times 1.0075$$

$$\therefore R = \frac{50000 \times 1.0075^{48} \times 0.0075}{1.0075^{48} - 1}$$

$$= \$1244 \text{ as reqd.}$$

$$\text{v) } \text{Owing} = A_{24} = 50000 \times 1.0075^{24} - \frac{1244(1.0075^{24} - 1)}{0.0075} \\ = \$27242$$

$$\text{vi) } 2200 = \frac{27242 \times 1.0075^n \times 0.0075}{1.0075^n - 1} \text{ (from iv)}$$

$$2200 \times 1.0075^n - 2200 = 204.315 \times 1.0075^n$$

$$1995.685 \times 1.0075^n = 2200$$

$$\therefore 1.0075^n \div 1.10$$

$\therefore n \div 13$ months to pay off the loan.